

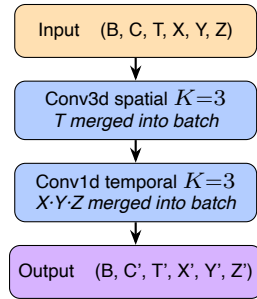
Patchify architecture — PatchEmbed3DPlus1D

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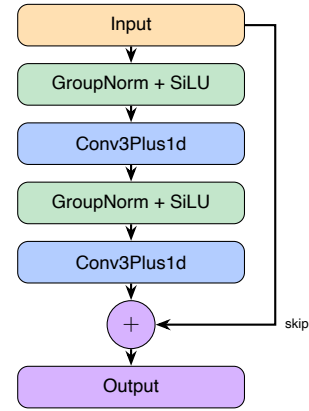
Hierarchical 3D+1D patchify encoder, inspired by MovieGen TAE (github.com/MathieuTuli/MovieGen, `tae.py:740`).

Building blocks

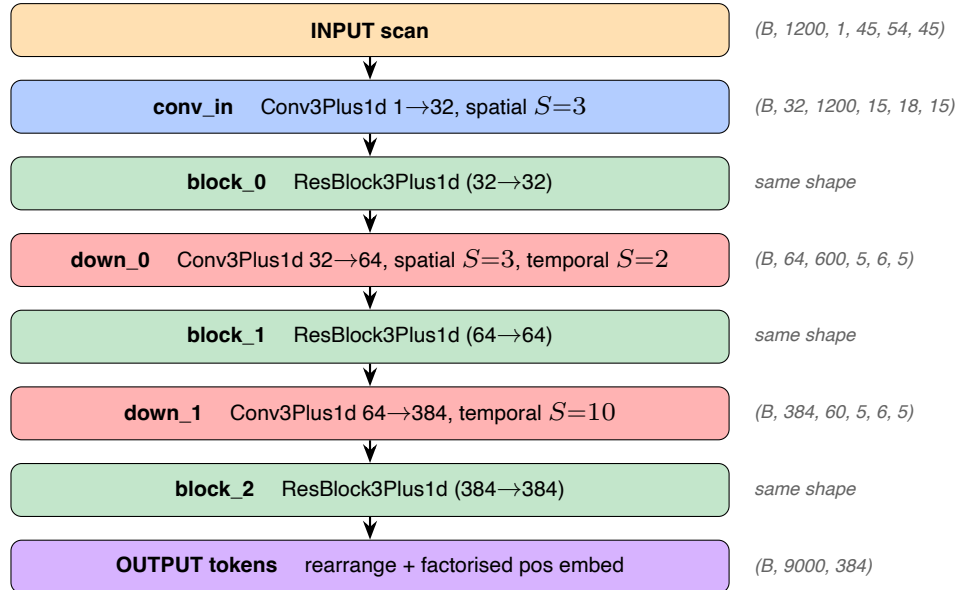
Conv3Plus1d — separable substitute for the missing PyTorch `Conv4d`. Splits a 4D conv into a 3D spatial conv (per frame) followed by a 1D temporal conv (per voxel-position).



ResBlock3Plus1d — residual block of 2 `Conv3Plus1d` with `GroupNorm + SiLU` pre-activation, output added to a skip connection. Same shape in / out.



Full encoder (HCP-YA setup, T = 1200, temporal_kernel = 20)



Parameter breakdown

Layer	What it contains	Params
conv_in	1 Conv3Plus1d (1{→}32)	4 000
block_0	2 Conv3Plus1d (32{→}32) + 2 GroupNorm	61 696
down_0	1 Conv3Plus1d (32{→}64)	67 712
block_1	2 Conv3Plus1d (64{→}64) + 2 GroupNorm	246 272
down_1	1 Conv3Plus1d (64{→}384)	2 138 880
block_2	2 Conv3Plus1d (384{→}384) + 2 GroupNorm	8 850 432
PositionEmbedding3D	pos_temporal + pos_spatial + pos_cls	81 024
TOTAL		11 450 016

Token grid output : $T_{\text{eff}} = 60$, $N_{\text{spatial}} = 5 \cdot 6 \cdot 5 = 150$, total **9 000 tokens** of dim 384.